

Name: _____

Date: _____



Year 10A Mathematics Course

Test 5 , 2015

Topics – Logs, Exponentials and Finance

 50

= _____ %

Total Time: 60 minutes
Total Reading: 5 minutes
Total Working: 55 minutes
Weighting: 5 % of the year.
Equipment Allowed: 1 page notes (A4 both sides), Scientific calculator

SECTION 1: CALCULATOR FREE

Time: 24 minutes	Marks for Section 1: 22
Reading: 2 minutes	Equipment Allowed: Nil
Working: 22 minutes	

1. [6 marks: 1, 1, 1, 1, 1, 1]

Complete the logarithm laws below:

a) $\log ab = \log a + \log b$ ✓

b) $\log \frac{a}{b} = \log a - \log b$ ✓

c) $\log a^n = n \log a$ ✓

d) $\log 1 = 0$ ✓

e) $\log 10 = 1$ ✓

f) $\log \frac{1}{a} = -\log a$ ✓

2. [2 marks: 1, 1]

Calculate:

a) $\log 100000 = \log 10^5$
 $= 5 \log 10$
 $= 5 \times 1$
 $= 5$
 ✓

b) $\log \sqrt[5]{10} = \log 10^{\frac{1}{5}}$
 $= \frac{1}{5} \log 10$
 $= \frac{1}{5}$
 ✓

3. [6 marks: 2, 4]

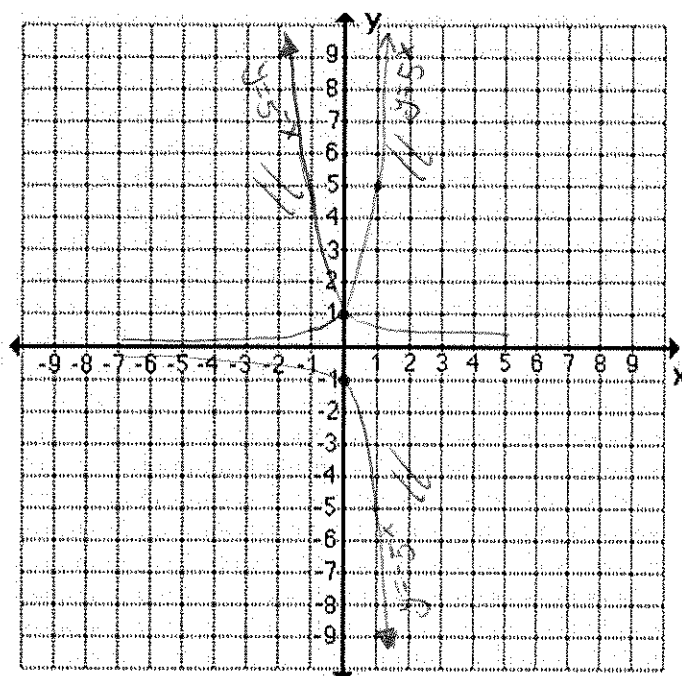
Simplify:

a) $\log_2 12 - \log_2 6 = \log_2 \frac{12}{6}$ ✓
 $= \log_2 2$
 $= 1$ ✓

b) $\log_4 12 + 3 \log_4 2 - \log_4 6 = \log_4 \frac{12 \times 2^3}{6}$ ✓✓
 $= \log_4 16$
 $= \log_4 4^2$ ✓
 $= 2 \log_4 4$
 $= 2$ ✓

4. [8 marks: 2, 4, 1, 1]

a) Graph the function $y = 5^x$. Hint: make table of values to calculate x and y coordinates of points on graph.



b) On the same coordinated plane, graph functions $y = -5^x$ and $y = 5^{-x}$.

c) Describe transformations between $y = 5^x$ and $y = -5^x$.

REFLECTION AROUND X-AXIS ✓

d) Describe transformations between $y = 5^x$ and $y = 5^{-x}$.

REFLECTION AROUND y-AXIS ✓

~END OF SECTION 1~

SECTION 2: CALCULATOR ASSUMED

Time:	36 minutes	Marks for Section 1:	28
Reading:	3 minutes	Equipment Allowed:	Nil
Working:	33 minutes		

5.	[8 marks: 2, 4, 2]		
	Sam wants to borrow \$4560 from a bank to buy a car. BankWest is charging him 8% interest per annum and he needs to pay it off in 5 years.		
a)	Calculate the simple interest he would pay during the time of the loan. $SI = 4560 \times \frac{8}{100} \times 5 = 1824$		
b)	Another bank, Westpac bank, is offering him the same conditions (to borrow \$4560 and pay it off in 5 years). The bank says that the total simple interest he would pay is \$1596. What is the interest per annum the bank is charging? $1596 = 4560 \times X \times 5$ $X = \frac{1596}{4560 \times 5}$ $X = 0.07$ BANK IS CHARGING 7% PER ANNUM		
c)	Which bank should Sam choose? Explain why. WESTPAC AS HE IS GOING TO PAY LESS INTEREST OVERALL		
6.	[6 marks: 3, 3]		
	Calculate how much interest (to the nearest cent) is added over 5 years to the account paying 8% p.a. interest on an initial sum of \$34000 if the interest is compounded:		
a)	monthly $CI = 34000 \left(1 + 0.08 \div 12\right)^{60}$ $CI = 50654.75$ $\begin{array}{r} 50654.75 \\ - 34000.00 \\ \hline 16654.75 \end{array}$ IT'S ADDED \$16654.75.		

b) half yearly

$$A = 34000 (1 + 0.08 \div 2)^{10} \checkmark$$

$$A = 50328.31 \checkmark$$

$$\begin{array}{r} 50328.31 \\ - 34000.00 \\ \hline 16328.31 \end{array} \quad \text{I'r's ADDED } \$16328.31$$

7. [6 marks: 2, 4]

Solve the following equations:

a) $2^{4x-5} = 8$

$$2^{4x-5} = 2^3$$

$$4x-5 = 3 \checkmark$$

$$4x = 8$$

$$x = 2 \checkmark$$

b) $20 \times 0.935^x = 52$

$$0.935^x = 2.6 \checkmark$$

$$\log 0.935^x = \log 2.6 \checkmark$$

$$x \log 0.935 = \log 2.6$$

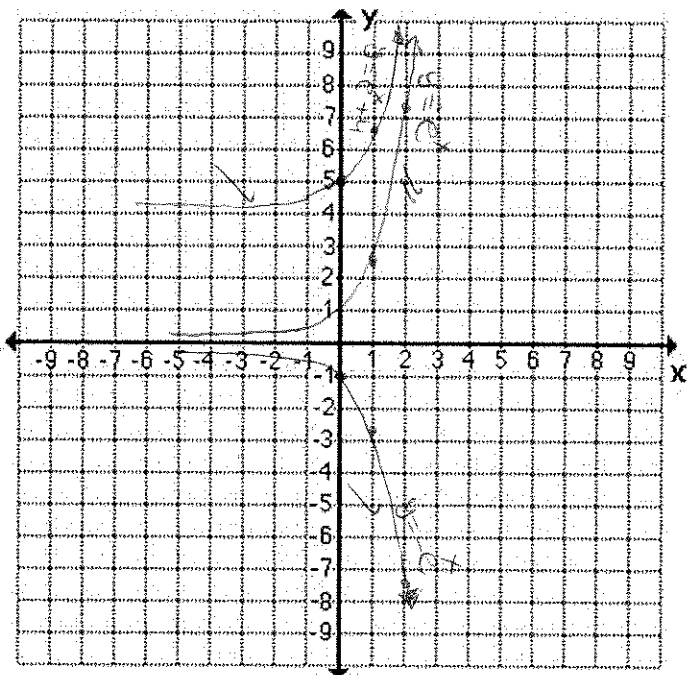
$$x = \frac{\log 2.6}{\log 0.935} \checkmark$$

$$x = -14.22 \checkmark$$

8. [8 marks: 4, 2, 2]

a) Graph the function $y = e^x$ by first completing a table of values below.

x	-2	-1	0	1	2
y	0.1353	0.3679	1	2.7183	7.3891



b)	Sketch the function and write its equation after $y = e^x$ has been translated 4 units up. $y = e^x + 4 \quad \checkmark$
c)	Sketch the function and write its equation after $y = e^x$ has been reflected about the x-axis. $y = -e^x \quad \checkmark$

~END OF SECTION 2~